GavinKlorfine–Assign2

Gavin Klorfine

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DDAR Exercise 2.3

The data set `UCBAdmissions` is a 3-way table of frequencies classified by `Admit`, `Gender`, and `Dept`.

(a) *Find the total number of cases contained in this table.*

```
data("UCBAdmissions")

sum(UCBAdmissions)
```

```
## [1] 4526
```

(b) *For each department, find the total number of applicants*

```
ucb <- as.data.frame(UCBAdmissions)

totals <- (
  ucb |>
  group_by(Dept) |>
  summarize(s = sum(Freq))
)

totals
```

```
## # A tibble: 6 x 2
##   Dept      s
##   <fct> <dbl>
## 1 A       933
## 2 B       585
## 3 C       918
## 4 D       792
## 5 E       584
## 6 F       714
```

(c) *For each department, find the overall proportion of applicants who were admitted*

```
admsns <- (
  ucb |>
  group_by(Dept, Admit) |>
  filter(Admit == "Admitted") |>
```

```r
  summarize(admit_freq = sum(Freq))
)

admsns$total_freq <- totals$s
admsns$ovr_prop <- admsns$admit_freq / admsns$total_freq
admsns |> select(Dept, ovr_prop)
```

```
## # A tibble: 6 x 2
## # Groups:   Dept [6]
##   Dept  ovr_prop
##   <fct>    <dbl>
## 1 A       0.644
## 2 B       0.632
## 3 C       0.351
## 4 D       0.340
## 5 E       0.252
## 6 F       0.0644
```

(d) *Construct a tabular display of department (rows) and gender (columns) showing the proportion of applicants in each cell who were admitted relative to the total applicants in that cell*

```r
berk_tab <- (
  prop.table(UCBAdmissions, c("Dept", "Gender"))
)

berk_tab <- berk_tab["Admitted",,] |> t() |> xtable()
caption(berk_tab) <- "Proportion of admitted applicants by department and gender using the UCBAdmissions
berk_tab
```

% latex table generated in R 4.5.1 by xtable 1.8-4 package % Sat Jan 24 20:30:41 2026

	Male	Female
A	0.62	0.82
B	0.63	0.68
C	0.37	0.34
D	0.33	0.35
E	0.28	0.24
F	0.06	0.07

Table 1: Proportion of admitted applicants by department and gender using the UCBAdmissions dataset.

DDAR Exercise 3.1

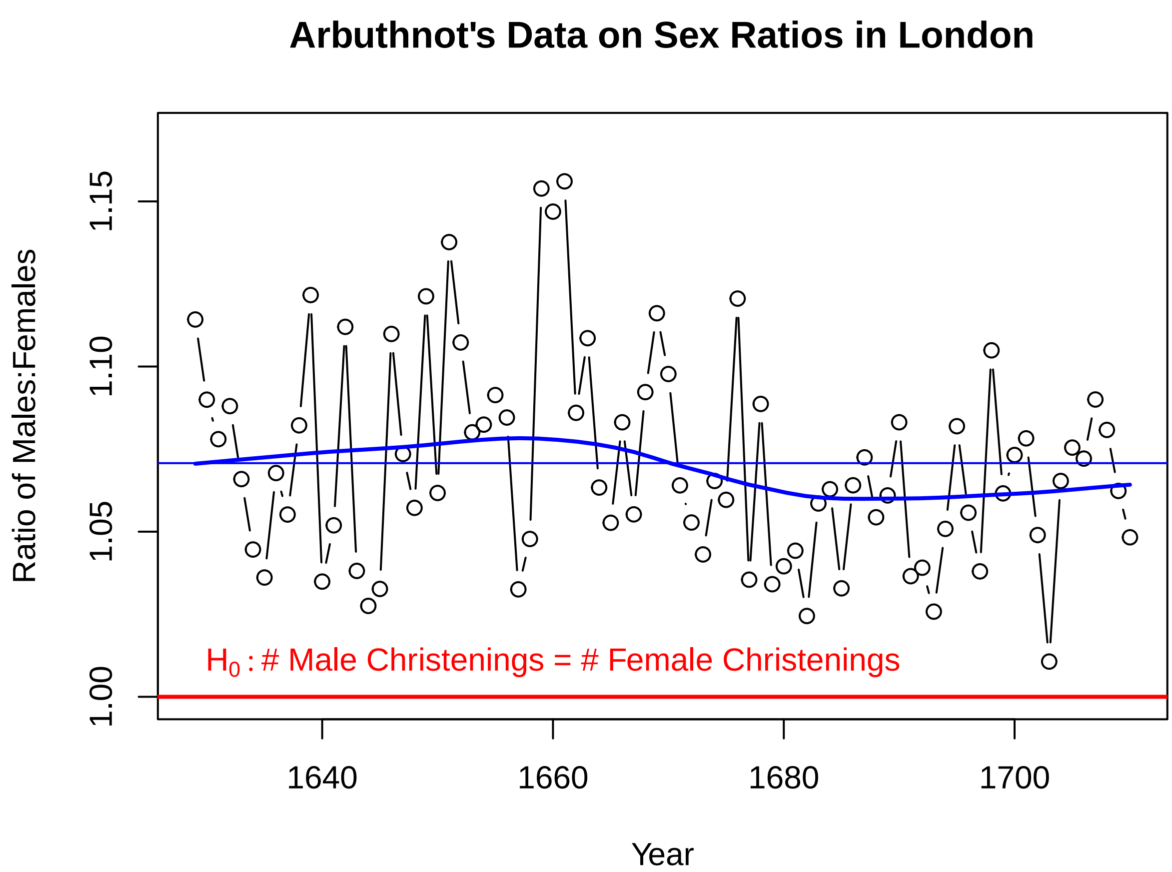
The `Arbuthnot` data in HistData (Example 3.1) also contains the variable `Ratio`, giving the ratio of male to female births.

(a) *Make a plot of `Ratio` over `Year`, similar to Figure 3.1. What features stand out? Which plot do you prefer to display the tendency for more male births?*

```r
# Code adapted from Friendly & Meyer (2016; p. 67), Discrete Data Analysis with R

data("Arbuthnot", package= "HistData")

with(Arbuthnot, {
   plot(x = Year, y = Ratio, type= "b",
   ylim= c(1.0, 1.17), main = "Arbuthnot's Data on Sex Ratios in London",
   ylab= "Ratio of Males:Females")
   abline(h= 1.0, col= "red", lwd= 2)
   abline(h= mean(Ratio), col= "blue")
   lines(loess.smooth(Year, Ratio), col= "blue", lwd= 2)
   text(x = 1660, y= 1.0, expression(H[0]: "# Male Christenings = # Female Christenings"),
   pos= 3, col= "red")
})
```



Features that stand out include no points dipping below 1.0, the value expected if there were an equal number of males and females christened. The average ratio of males:females christened from 1629 to 1710 was 1.071; this average ratio tended to be larger from 1629 to ~1670 (ratio = 1.082), whereas it tended to be smaller from ~1670 to 1710 (ratio = 1.059). Despite these fluctuations, the ratio remained near the mean, indicating a consistent trend of more male births than female births.

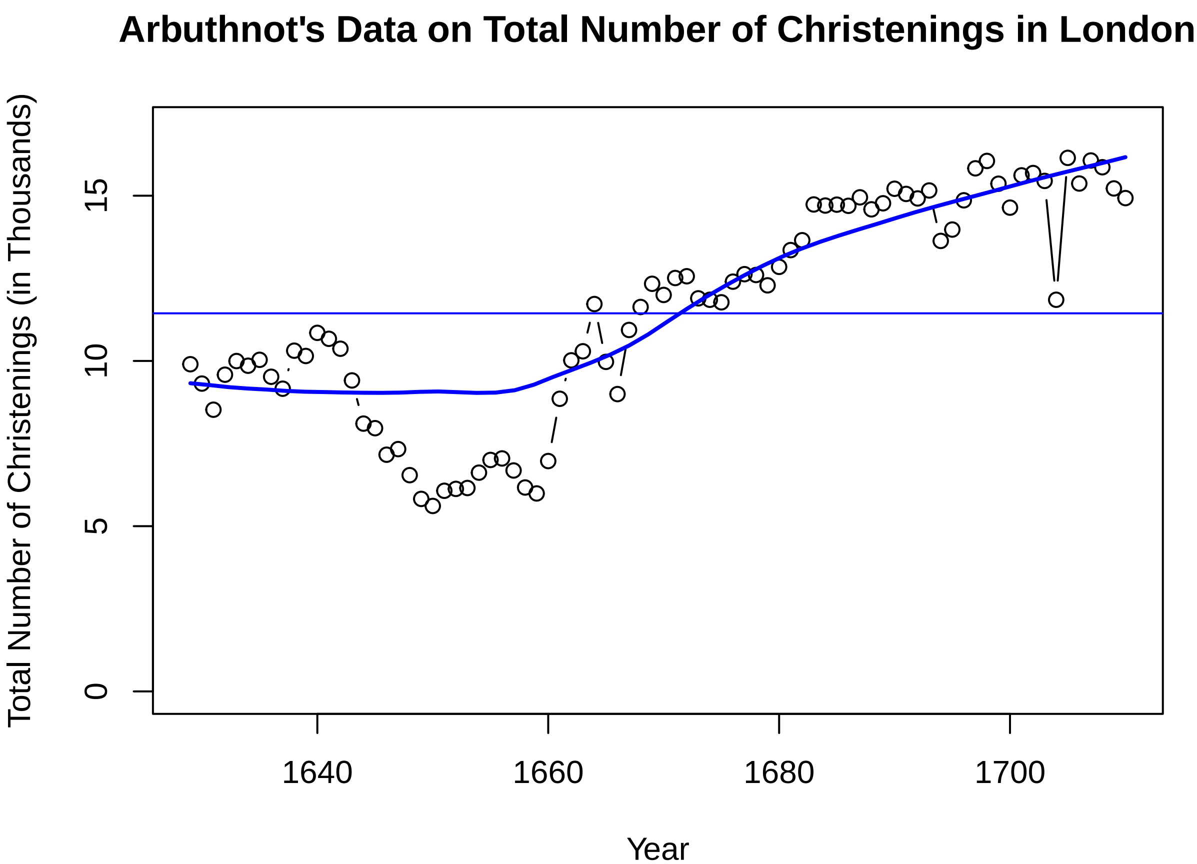
I prefer the plot in Figure 3.1 as it better represents the underlying process (i.e., births as Bernoulli trials where there is a probability that the child will be male). Probabilities rather than ratios also allow for modelling via a binomial distribution.

(b) *Plot the total number of christenings, Males + Females or Total (in 000s) over time. What unusual features do you see?*

```
# Code adapted from Friendly & Meyer (2016; p. 67), Discrete Data Analysis with R

data("Arbuthnot", package= "HistData")

with(Arbuthnot, {
   plot(x = Year, y = Total, type= "b",
   ylim= c(0, 17), main = "Arbuthnot's Data on Total Number of Christenings in London",
   ylab= "Total Number of Christenings (in Thousands)")
   abline(h= mean(Total), col= "blue")
   lines(loess.smooth(Year, Total), col= "blue", lwd= 2)
})
```



An unusual feature is that of the data between ~1645 and 1660, as they form a trough in what would otherwise be a steady increase in total christenings. Furthermore, the descent into the trough is somewhat gradual, occurring over ~10 years, whereas the ascent from the trough is quick and occurs over ~5 years.

There is also a data point near ~1705 that seems atypical, as the number of christenings is a few thousand less than the surrounding data points (i.e., 12,000 as compared to a typical value of 15,000 for the previous 20 years).

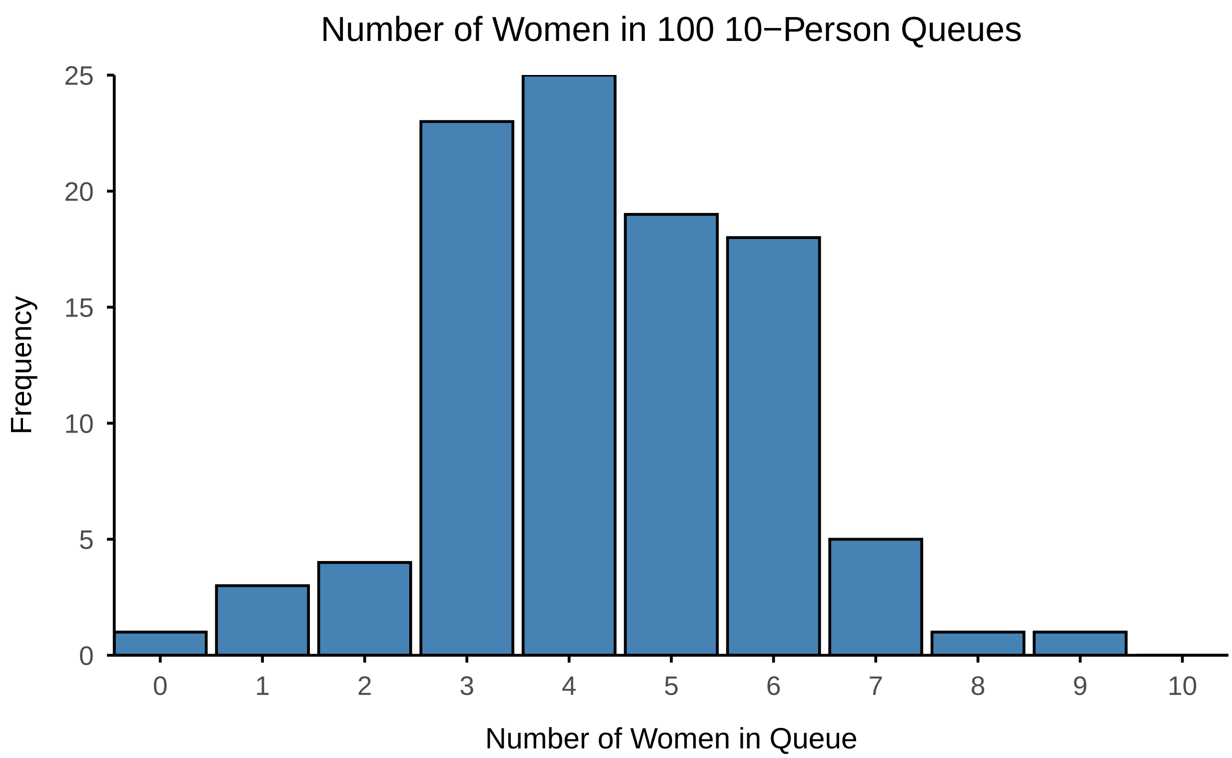
DDAR Exercise 3.3

Use the data set `WomenQueue` to:

(a) *Produce plots analogous to those shown in Section 3.1 (some sort of bar graph of frequencies).*

```
data("WomenQueue")
wq <- as.data.frame(WomenQueue)

ggplot(wq, aes(x = nWomen, y = Freq)) +
  geom_bar(
    stat ="sum",
    fill = "steelblue",
    color = "black"
  ) +
  scale_y_continuous(expand = c(0, 0)) +
  scale_x_discrete(expand = c(0, 0)) +
  labs(
    title = "Number of Women in 100 10-Person Queues",
    x = "Number of Women in Queue",
    y = "Frequency"
  ) +
  gavinTheme()
```

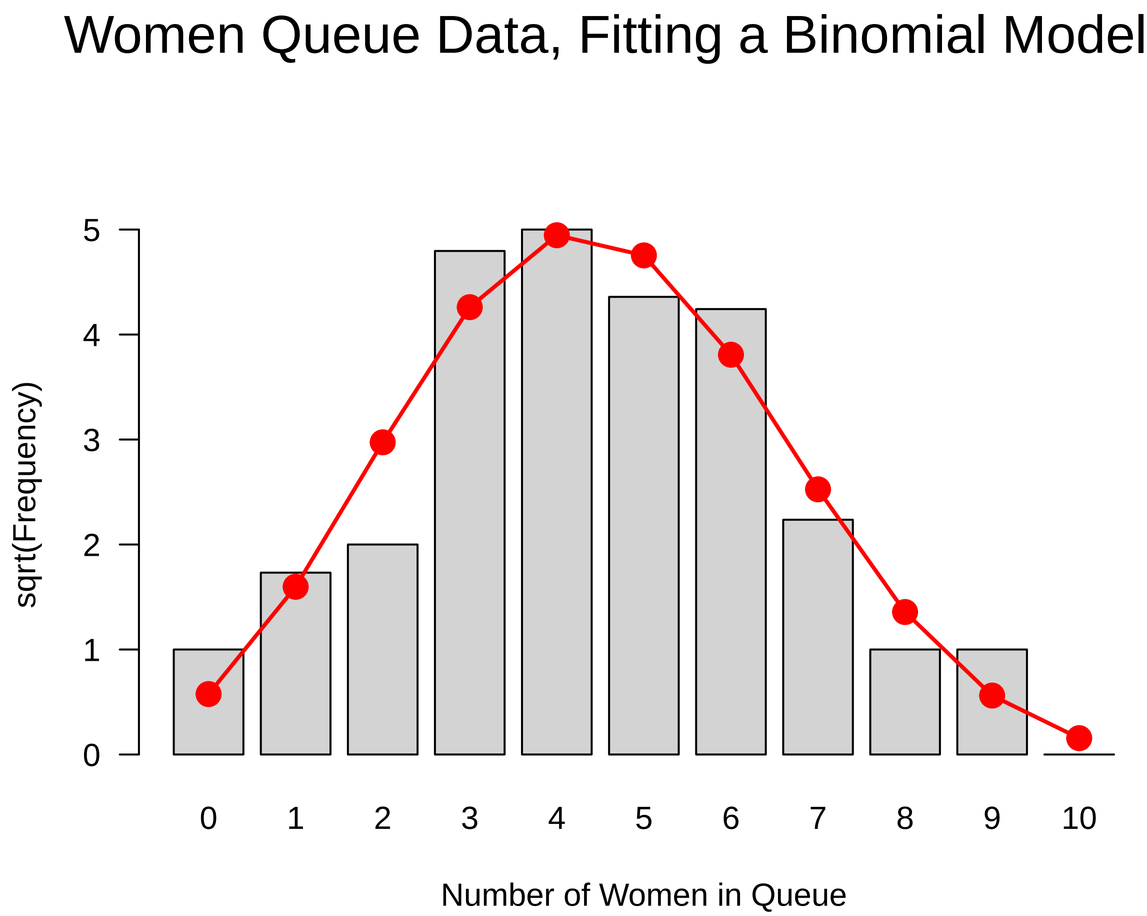


(b) *Check for goodness-of-fit to the binomial distribution using the `goodfit()` methods described in Section 3.3.2.*

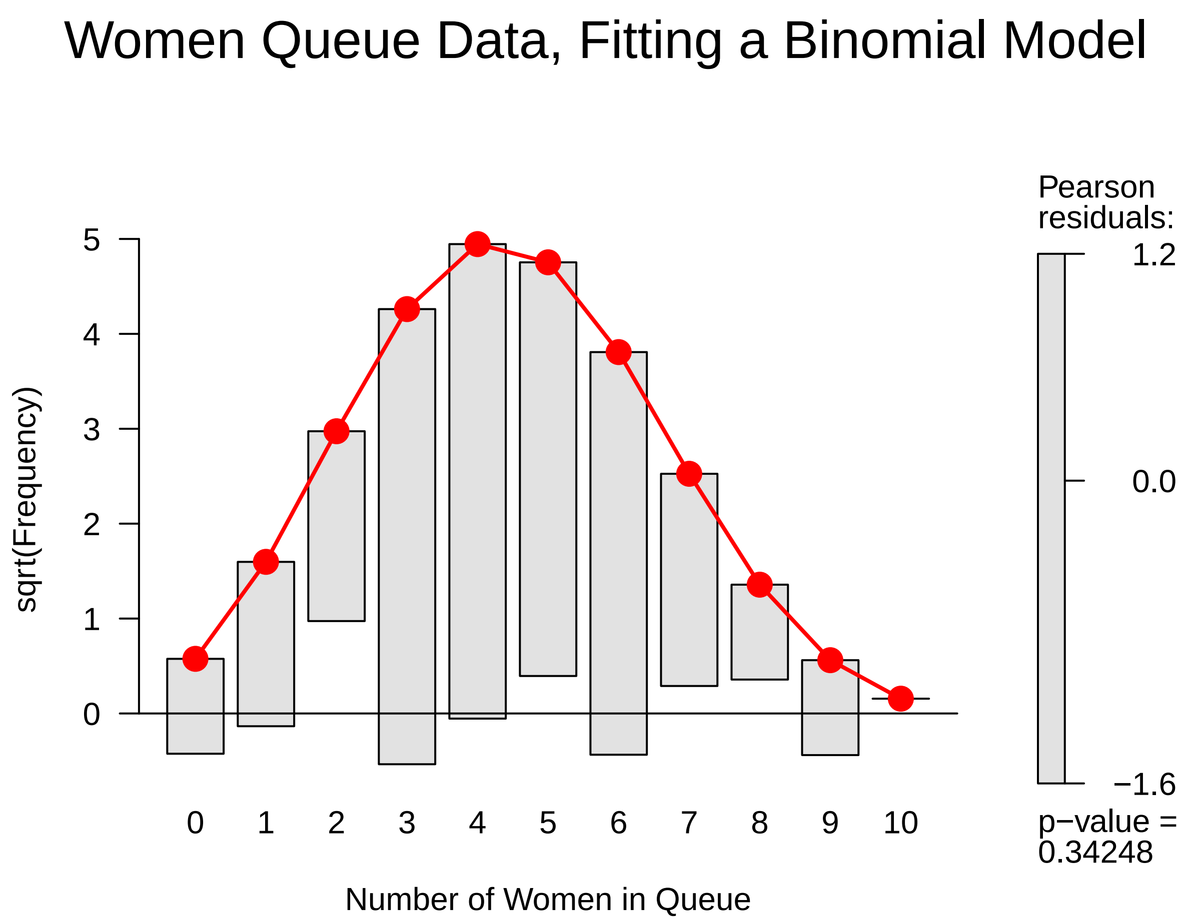
```
wq_fit <- goodfit(WomenQueue, type = "binomial")
summary(wq_fit)
```

```
##
##   Goodness-of-fit test for binomial distribution
##
##                       X^2 df  P(> X^2)
## Likelihood Ratio 8.650999  8 0.3725869
```

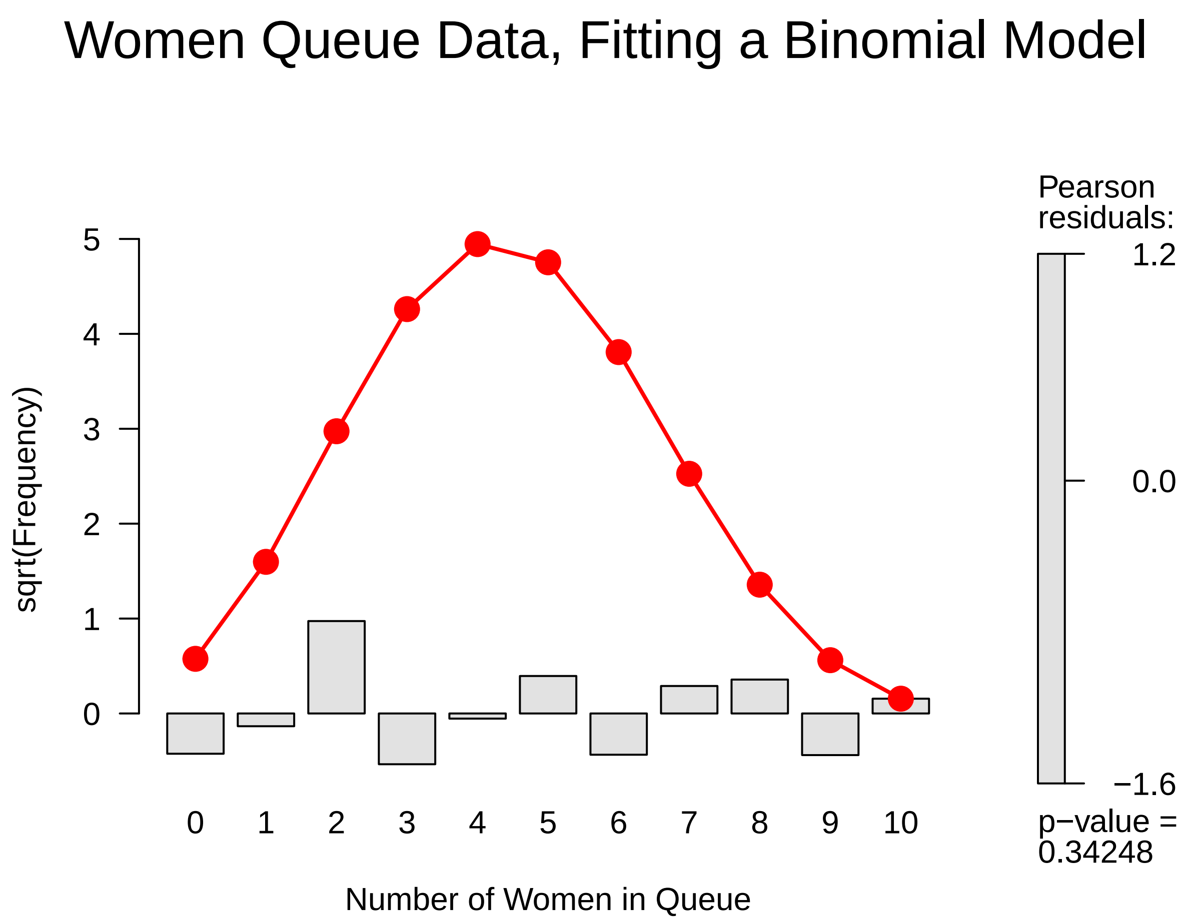
```
plot(wq_fit, type="standing",
     main = "Women Queue Data, Fitting a Binomial Model",
     xlab = "Number of Women in Queue"
)
```




```
plot(wq_fit, type="hanging", shade = TRUE,
     main = "Women Queue Data, Fitting a Binomial Model",
     xlab = "Number of Women in Queue"
)
```



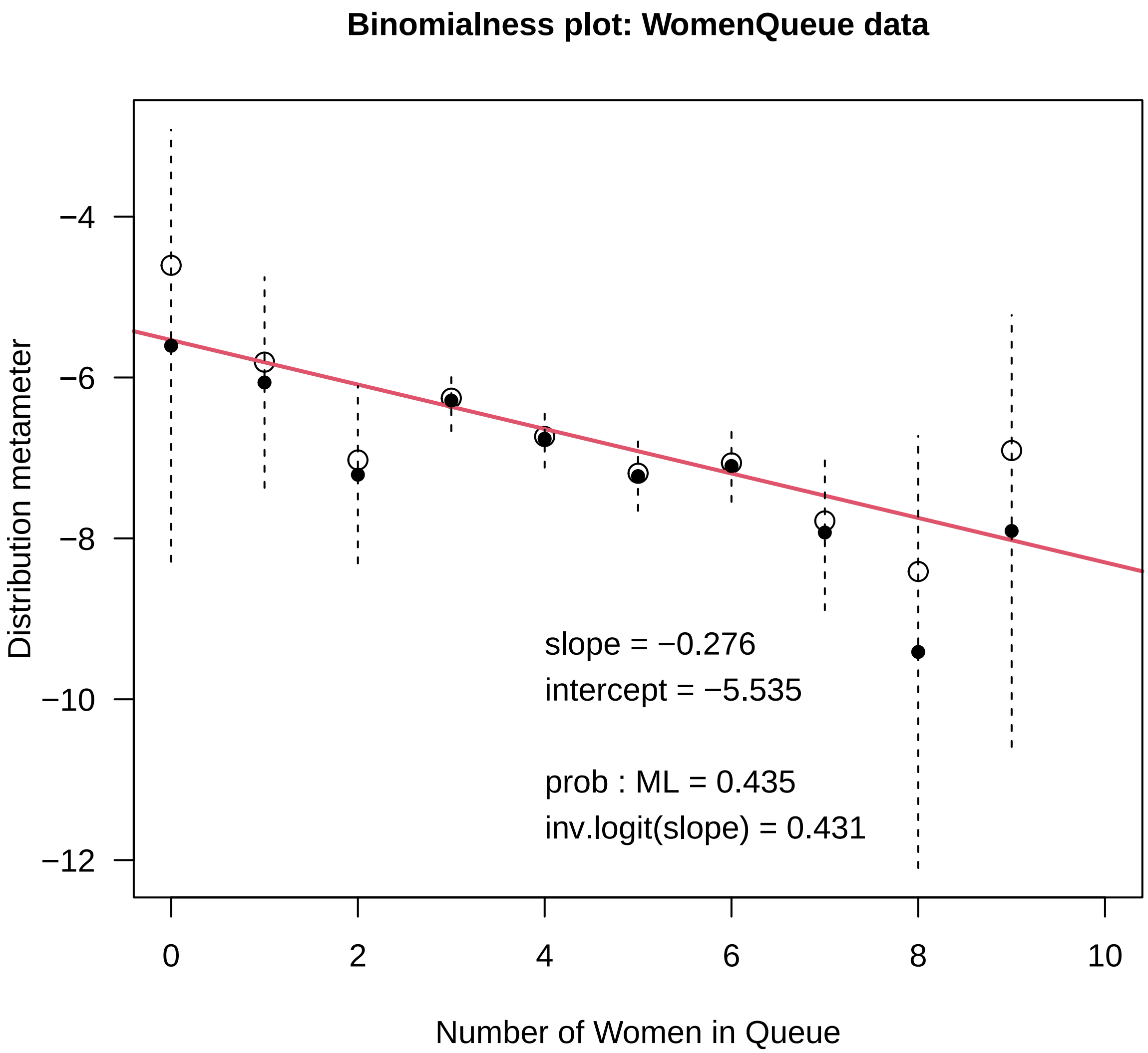

```r
plot(wq_fit, type="deviation", shade = TRUE,
     main = "Women Queue Data, Fitting a Binomial Model",
     xlab = "Number of Women in Queue"
)
```



Due to a non-significant p value and no residuals exceeding ± 2, $\text{Bin}(10, 0.435)$ can be considered a good fit for the data.

(c) *Make a reasonable plot showing departure from the binomial distribution.*

```
distplot(
  WomenQueue,
  type = "binomial",
  xlab = "Number of Women in Queue",
  main = "Binomialness plot: WomenQueue data"
)
```



(d) *Suggest some reasons why the number of women in queues of length 10 might depart from a binomial distribution,* $\text{Bin}(n = 10, p = 1/2)$

Because the true probability of selecting a woman from a queue of 10 may differ from $p = 1/2$ (in the place and time data was collected). Perhaps there were more men taking the subway or there were more men in London, generally.

DDAR Exercise 3.4

Continue Example 3.13 on the distribution of male children in families in Saxony by fitting a binomial distribution, $\text{Bin}(n = 12, p = \frac{1}{2})$*, specifying equal probability for boys and girls. [Hint: you need to specify both `size` and `prob` values for `goodfit()`.]*

(a) *Carry out the GOF test for this fixed binomial distribution. What is the ratio of* χ^2/df*? What do you conclude?*

```r
data("Saxony")

# GOF test
sax_fit <- goodfit(
  Saxony,
  type = "binomial",
  par = list(size = 12, prob = 0.5)
)
sum_sax <- summary(sax_fit)
```

```
##
##   Goodness-of-fit test for binomial distribution
##
##                         X^2 df     P(> X^2)
## Pearson          249.1954 12 2.013281e-46
## Likelihood Ratio 205.4060 12 2.493625e-37
```

```r
# Ratio of chisq/df
(chi_df <- sum_sax["Pearson", "X^2"] / sum_sax["Pearson", "df"])
```

```
## [1] 20.76629
```

I conclude that $\text{Bin}(n = 12, p = \frac{1}{2})$ is a significantly bad fit to the `Saxony` data, $\chi^2(12) = 249.195$, $p < .001$. Furthermore, the ratio of $\chi^2/df = 20.766$ is substantially greater than 1, which would be the expected value under the null hypothesis of acceptable fit.

(b) *Test the additional lack of fit for the model* $\text{Bin}(n = 12, p = \frac{1}{2})$ *compared to the model* $\text{Bin}(n = 12, p = \hat{p})$ *where* $\hat{p}$ *is estimated from the data.*

```r
sax_fit_phat <- goodfit(
  Saxony,
  type = "binomial",
  par = list(size = 12)
)
sum_sax_phat <- summary(sax_fit_phat)
```

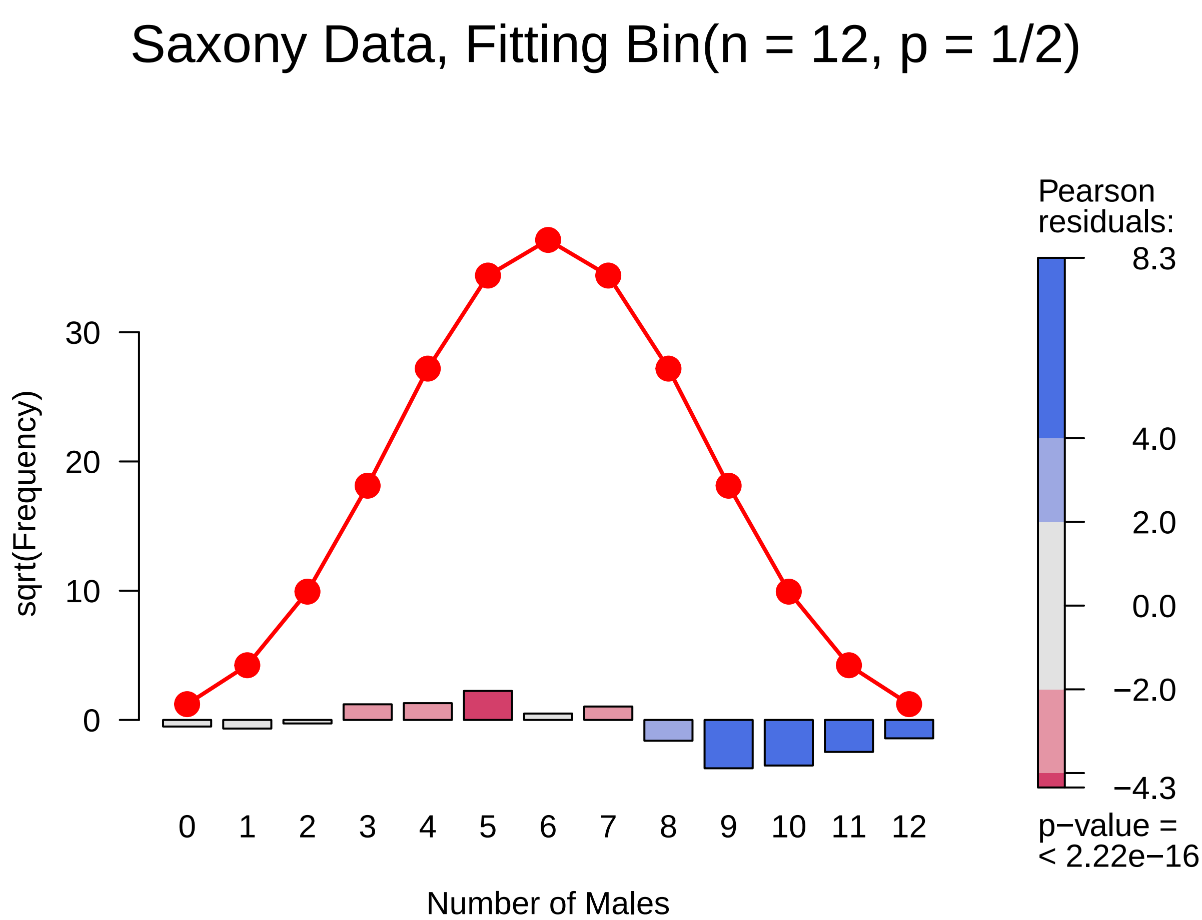
```
##
##   Goodness-of-fit test for binomial distribution
##
##                        X^2 df     P(> X^2)
## Likelihood Ratio 97.0065 11 6.978187e-16
```

```r
phat <- unlist(sax_fit_phat$par)["prob"]
```

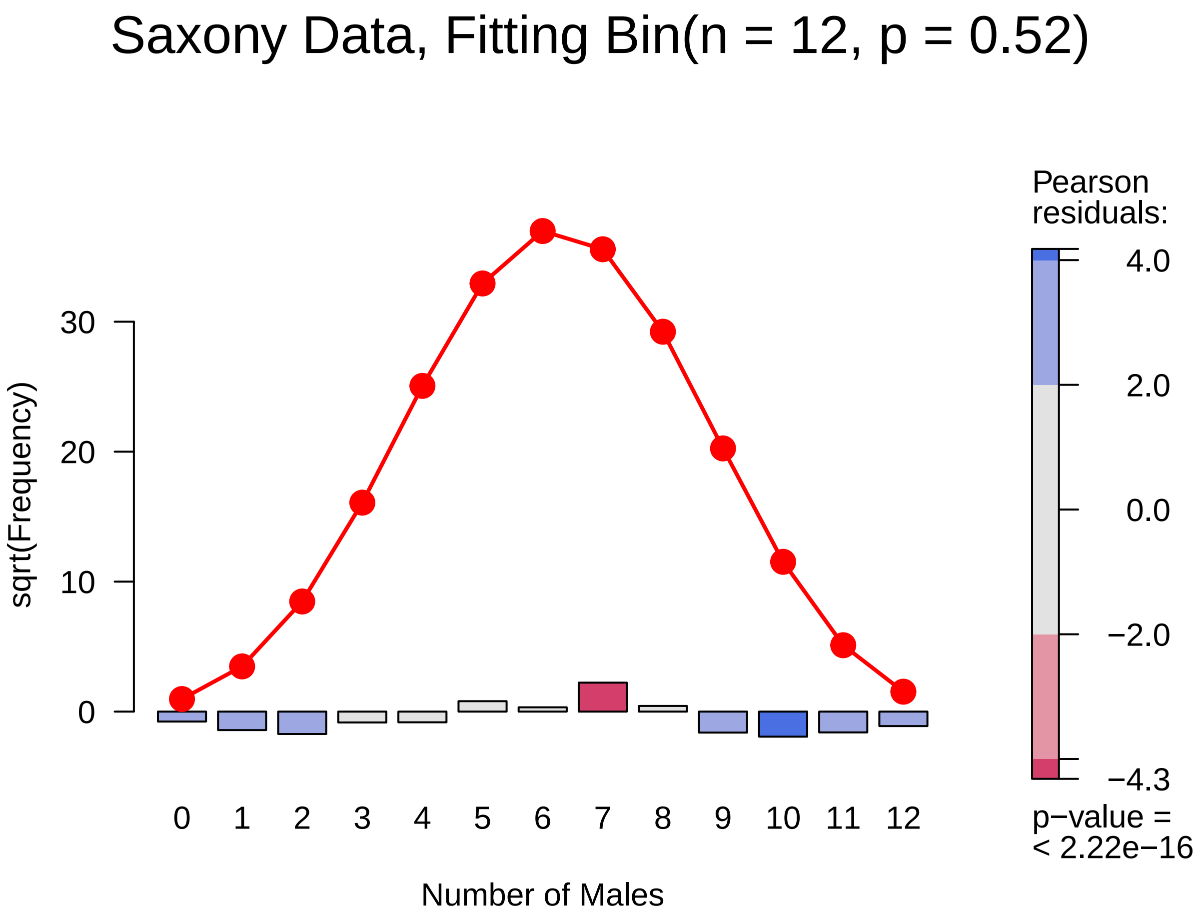
While the likelihood ratio improved (i.e., lowered) by estimating $\hat{p}$ from the data ($G^2_{\hat{p}=0.519}(11) = 97.01$ versus $G^2_{p=0.5}(12) = 205.41$), there was still a significant departure from the null hypothesis of good fit ($p < .001$), meaning that $\text{Bin}(n = 12, \hat{p} = 0.519)$ is still a very bad fit for the `Saxony` data.

(c) *Use the `plot.goodfit()` method to visualize these two models.*

```r
plot(
  sax_fit,
  type = "deviation",
  shade = TRUE,
  main = "Saxony Data, Fitting Bin(n = 12, p = 1/2)",
  xlab = "Number of Males"
)   # 3.4a
```




```r
plot(
  sax_fit_phat,
  type = "deviation",
  shade = TRUE,
  main = paste0("Saxony Data, Fitting Bin(n = 12, p = ", round(phat, 2), ")"),
  xlab = "Number of Males"
)   # 3.4b
```


Saxony Data, Fitting Bin(n = 12, p = 0.52)
sqrt(Frequency)
0
10
20
30
0
1
2
3
4
5
6
7
8
9
10
11
12
Number of Males
Pearson
residuals:
4.0
2.0
0.0
−2.0
−4.3
p−value =
< 2.22e−16